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TCX2005: Information Systems, Management and Organisations

## L3: IS, Organizations, & Strategy-I

Prakash C. Sukhwal  
Senior Lecturer, DISA  
[pcsdisa@nus.edu.sg](mailto:pcsdisa@nus.edu.sg)

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# Lesson Objectives



Organization and IS: The Two-way Relationship



Which features of organizations do managers need to know about to build and use IT/IS successfully?



Impacts of IS on an Organization



How do Porter's competitive forces model help companies develop competitive strategies using information systems?

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# Learning Resources



- Textbook (read before class)
  - Read chapter 3
- Any Other Material
  - Shared on Canvas for this week

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# Case Study

Read the case study from textbook on –[N26: A Bank Without Branches](#)

Identify:

- a. How does N26's strategic and technical approach made them competitive in the industry?
- b. What are the key IS (technological) capabilities offered by N26?
- c. What business outcomes are achieved through strategic use of Information systems?

Note: be ready to share your thoughts

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# Case Study

## Strategic Approach:

- Mobile bank without physical branches
- Partnership model for physical services (Cash26 with retail partners)
- Customer-centric processes usable anywhere in the world
- Simple, fully online banking from identification to account opening

## Tech Approach:

- Cloud-based single system architecture (vs. traditional distributed IT)
- Over 100 microservices running in different software languages/frameworks
- Configurable software for geographic expansion (US, Brazil)
- Ability to open accounts within 8 minutes
- Digital card features for instant mobile wallet payments

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# Case Study

## Business Outcomes:

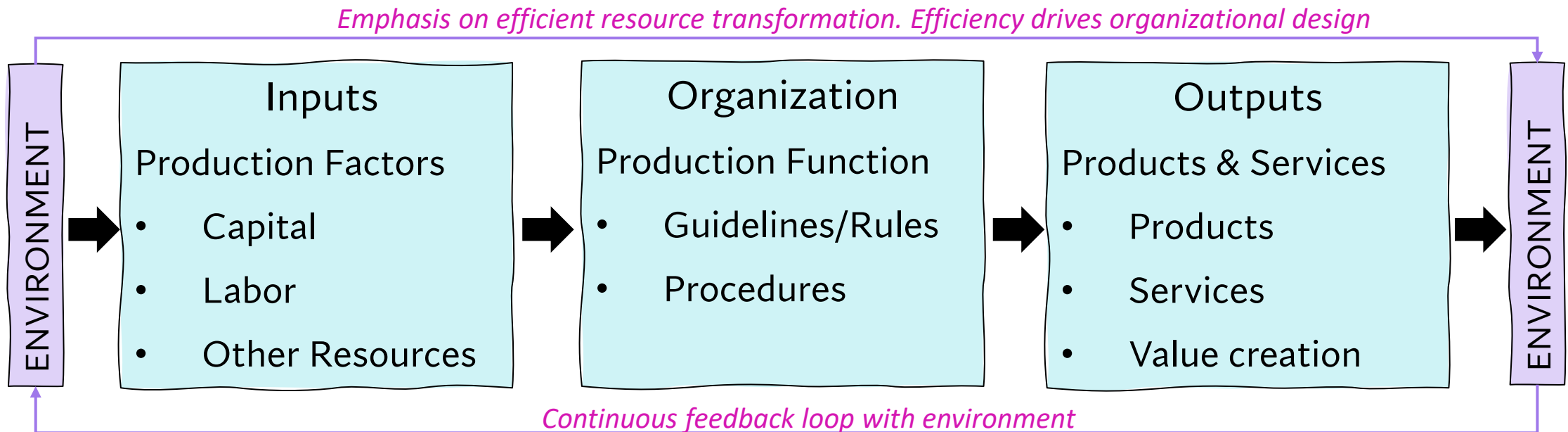
- Built a network of 5+ million customers in 24 European countries
- Successfully expanded to United States in partnership with Axos Bank
- Raised close to \$800 million from investors
- Achieved market value of \$3.5 billion (as of May 2020)
- Positioned as one of the most valuable fintech startups in Europe

## Learning:

- Modern technology stack creates competitive advantages
- Banking processes can be fully digital yet customer-centric
- Information systems help businesses compete in different markets
- Finding the right technology mix establishes unique market position

# Organization

- An organization transforms capital and labor into products and services through a production process.
- These outputs are then consumed by the environment, which in turn provides additional capital and labor as inputs



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# Organization

## Key Perspectives

### Technical View

- Resource transformation process
- Capital and labor utilization
- Production efficiency
- Economic outputs

### Behavioral View

- Rights and privileges
- Obligations and responsibilities
- Values and norms
- Working relationships

Implementing new information systems requires understanding both perspectives. Technical changes must be balanced with organizational changes in relationships, responsibilities, and established practices.



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# Example

## Singapore Airlines Picks Salesforce For Major Revamp Of Customer Case And Knowledge Management System

06 April 2021 - Singapore Airlines (SIA) has selected Salesforce, the global leader in customer relationship management (CRM), to develop and implement a new customer case and knowledge management system for the Airline.

This new system, 1Point, will be deployed across SIA's contact centres and customer feedback handling units globally from late 2021.

As a dynamic case management tool, 1Point offers a contextual and holistic view of customer interactions through a single interface. SIA's service agents will no longer need to toggle between multiple systems to retrieve customer data, up-to-date information on products, services, policies and procedures, and perform transactions. Its case management and guided workflows will also be integrated with SIA's in-house artificial intelligence and machine learning capabilities.

1Point will help to enhance the travel experience by supporting consistent and more personalised service across various ground-based customer touch points. Service agents will have access to the relevant information, including those related to each customer's previous interactions, at every step. This facilitates

When SIA implemented their customer service system, they:

1. Balanced the technical aspects (software development, data integration, AI tools)
2. Invested heavily in training and change management, acknowledging:
  - The importance of staff behavior and acceptance (Behavioral View)
  - That system success depends on how people use it, not just how well it works technically

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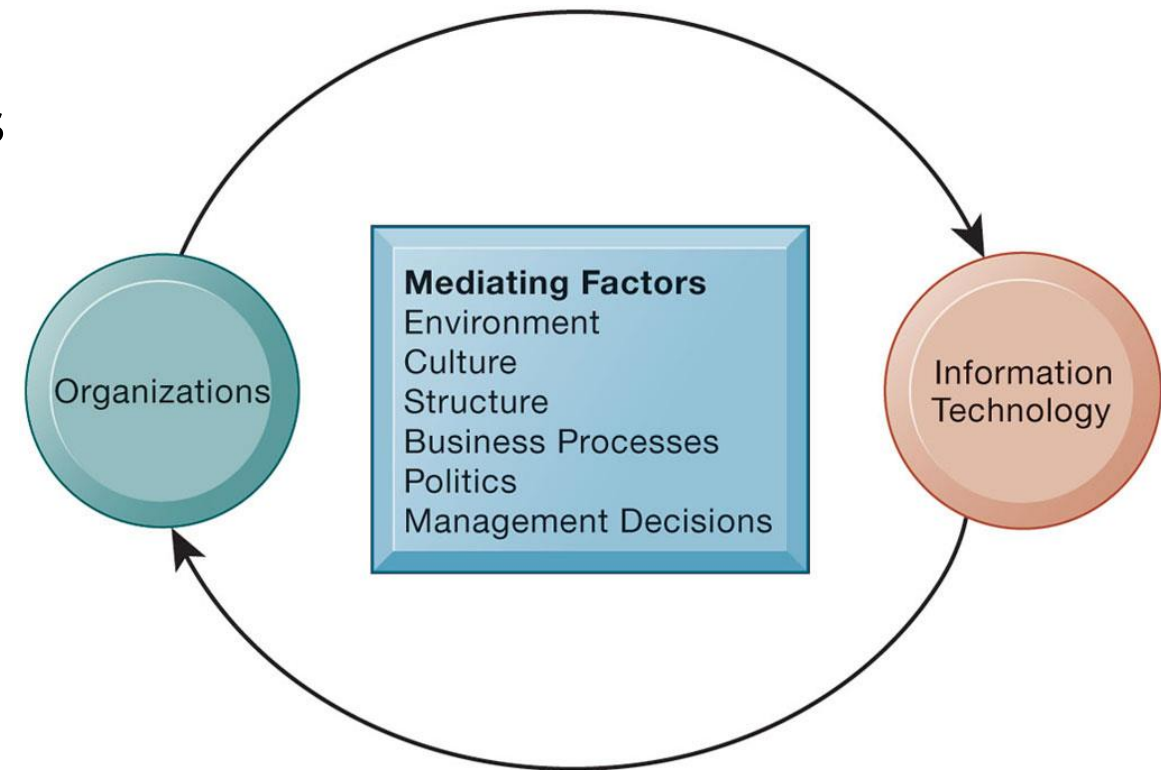
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# Organization and IS: The Two-Way Relationship

Information technology and organizations influence each other

- Relationship influenced by organization's
  - Structure
  - Business processes
  - Politics
  - Culture
  - Environment
  - Management decisions



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# Organization and IS: The Two-Way Relationship

## Organizations → IS

Information systems are built by teams to serve business interests and achieve organizational goals

## IS → Organizations

Organizations must adapt to benefit from new technologies and their capabilities

## Mediating Factors

Environment

Culture

Structure

Business  
Process

Politics

Management  
Decisions

# Example

## DBS's journey to becoming the best bank in the world through digital and cultural transformation

September 28, 2018



Since 2014, DBS's transformation has been nothing short of impressive. Nearly every aspect of the way it operates and how it interacts with customers has been digitalized.

Sponsored by



source

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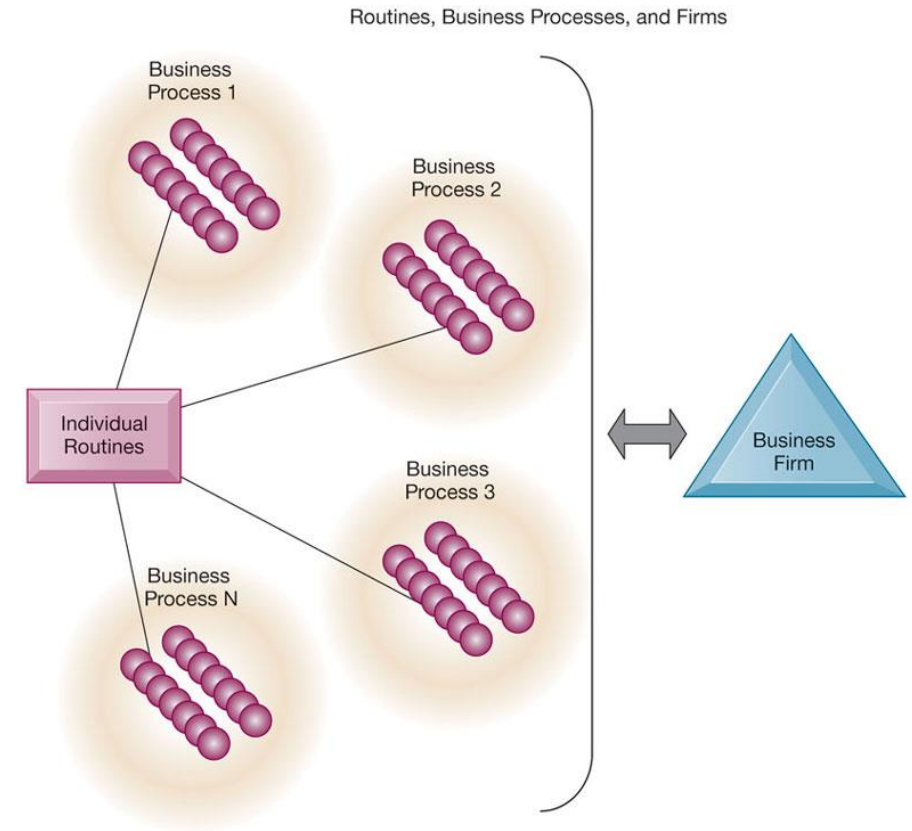
# Features of Organizations

- Bureaucracies with clear-cut divisions of labor and specialization
- Accountability, authority in system of impartial decision making
- Adherence to principle of efficiency
  - Maximizing output using limited inputs
- Routines and business processes
- Organizational politics, culture, structure, and environments

# Routines and business processes

Precise rules, procedures, and practices developed to cope with virtually all expected situations

- Business processes: Collections of routines
- Business firm: Collection of business processes



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# Politics and Culture

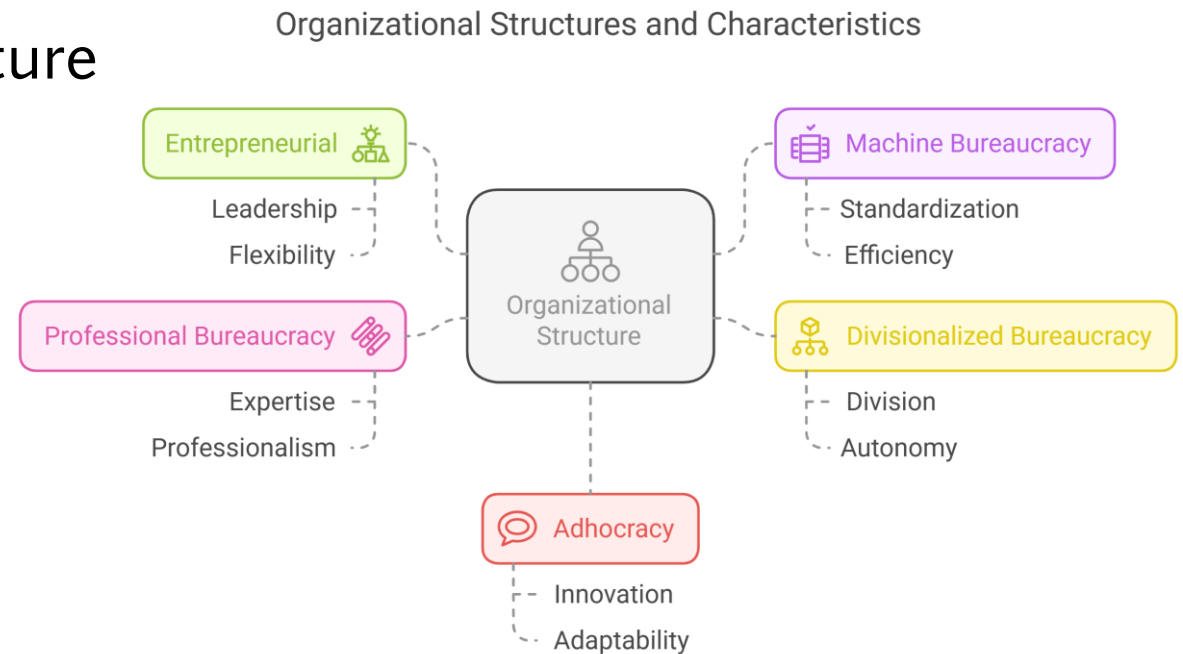
- Organizational politics
  - Divergent viewpoints lead to political struggle, competition, and conflict
  - Political resistance greatly hampers organizational change
- Organizational Culture
  - Encompasses set of assumptions that define goal and product
    - What products the organization should produce
    - How and where it should be produced
    - For whom the products should be produced
  - May be powerful unifying force as well as restraint on change

# Organizational Structure

The formal configuration between individuals and groups regarding the allocation of tasks, responsibilities, and authority within the organization

- Five basic kinds of organizational structure (Mintzberg, 1971)

- Entrepreneurial
- Machine bureaucracy
- Divisionalized bureaucracy
- Professional bureaucracy
- Adhocracy

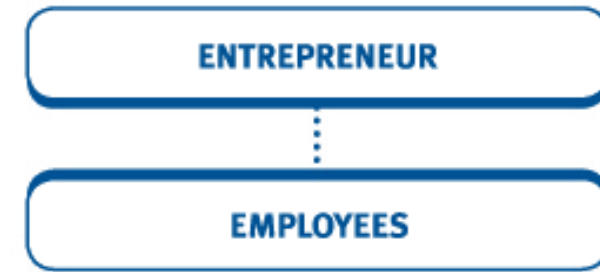




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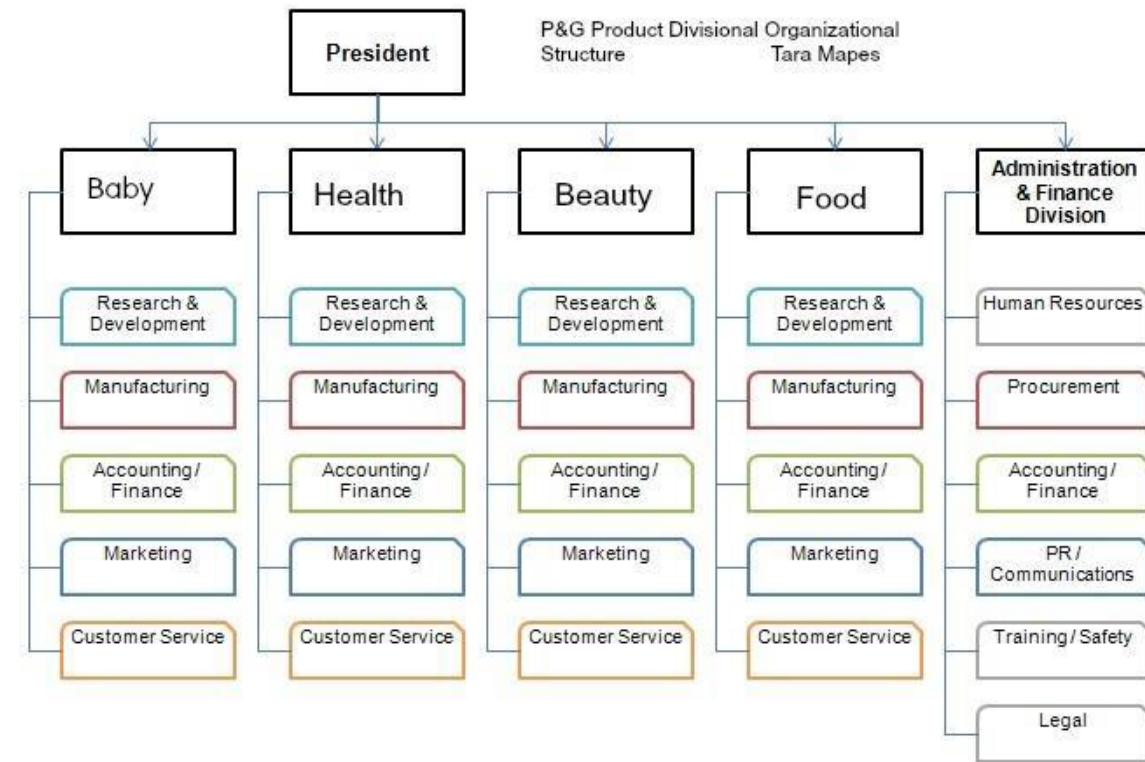
# Organizational Structure (cont.)

- Entrepreneurial
  - Simple structure
  - Fast-changing environment
  - Major decision made by one or two key personnel
  - Direct supervision
    - E.g., small start-up business
- Machine bureaucracy
  - Centralized management/decision making
  - Slowly changing environment -> stable environment
  - A high degree of formalization and work specialization
  - Standardization of work processes
  - Many levels exist in the chain of command from top management to the bottom of organization
  - Goal: achieve internal efficiency
    - E.g., midsize manufacturing firm, government agencies



# Organizational Structure (cont.)

- Divisionalized bureaucracy
  - Multiple machine bureaucracies, each producing a different product or service
  - Decision making is decentralized at divisional level
  - Little coordination among the separate divisions
  - Topped by central headquarter
    - E.g., Fortune 500 firms with multiple product lines



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# Organizational Structure (cont.)

- Professional bureaucracy
  - Knowledge-based organization
  - Standardization of skills
  - Goods and services depend on the expertise and knowledge of professionals
  - Highly trained professionals provide nonroutine services to clients
  - Relatively formalized but decentralized to give autonomy to professionals
  - Department heads with weak centralized authority
    - Top management is small, few middle managers
  - Goal: Innovate and provide high-quality services
    - E.g., law firms, universities, hospitals

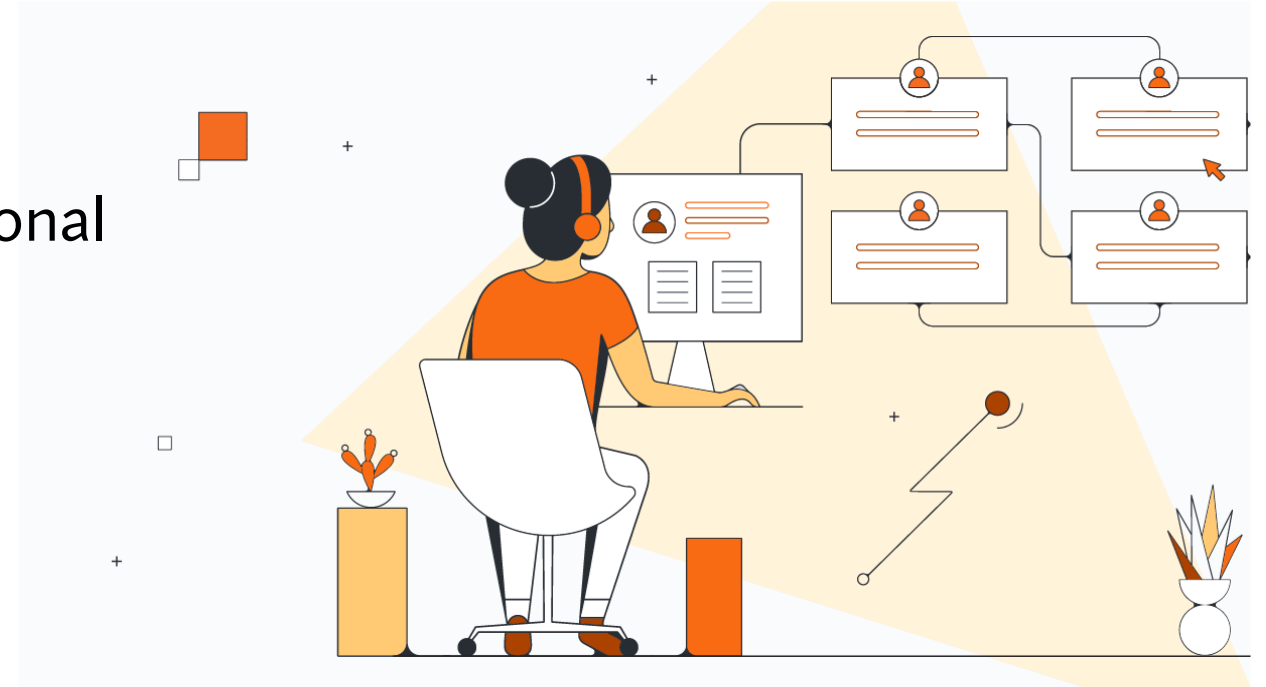
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# Organizational Structure (cont.)

- Adhocracy
  - Low in formalization and decentralization
  - Task force organization that must respond to rapidly changing environment
  - Engage in nonroutine tasks and use sophisticated technology
  - Goal: Innovation and rapid adaptation to changing environment
  - Large groups of specialists organized in short-lived multidisciplinary teams
  - Weak central management
    - E.g., research and development firms

# Organizational Structures and IT/IS

- Information system often reflects organizational structure
  - E.g., centralization/decentralization
- IT/IS could also influence organizational structure
- Organization structure influences IT/BA/Information Security/... unit performance



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# Organizational Environments

- Organizations and environments have a reciprocal relationship
- Organizations are open to, and dependent on, the social and physical environment
- Organizations can influence their environments
- Environments generally change faster than organizations
- Information systems can be instrument of environmental scanning, act as a lens
  - E.g., track competitor movements, customer sentiment, or market trends

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# Impact of IS on Organizations

Information systems have fundamentally transformed how organizations operate and make decisions in the modern business environment.

IS serve as the backbone of modern organizations, enabling real-time operations and strategic decision-making capabilities.

- **Strategic Integration:** Information systems have evolved from basic data processing tools to strategic assets that drive competitive advantage.
  - E.g., Amazon's sophisticated inventory management system allows it to maintain minimal stock while ensuring rapid delivery to customers.
- **Operational Excellence:** Organizations rely on IS for minute-to-minute operations, from processing customer orders to managing global supply chains.
  - E.g., Walmart uses its retail link system to connect with thousands of suppliers in real-time, automatically triggering reorders when inventory reaches certain levels.

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# Economic Impacts

**Three** key areas where IS transforms organizational economics

- **Cost transformation:** IS transforms organizational costs by enabling automation of manual processes and allowing more efficient resource allocation. Organizations can serve more customers with fewer costs.
  - E.g., banks have replaced many physical branches with online banking systems, reducing operational costs while improving service availability.
  - E.g., retailers like Amazon operate with fewer physical stores by using sophisticated inventory management systems that coordinate digital orders with automated warehouses.
- **Transaction Cost Reduction:** Modern IS reduces the costs of market participation.
  - E.g., Apple can efficiently manage global supply chains with partners like Foxconn through sophisticated information systems, making outsourcing more viable than vertical integration.



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## Economic Impacts (cont.)

- **Internal Management Cost Reduction:** IS reduces internal management costs by automating coordination and control functions that traditionally required layers of middle management.
  - E.g., retail chains no longer need multiple layers of managers to compile and report sales data - their systems automatically collect, analyze, and distribute this information.
  - E.g., Eastman Chemical Company leveraged information systems to increase revenue from \$3.3B to \$9.3B while reducing their workforce from 24,000 to 14,500 employees, particularly in middle management positions.

# Organizational & Behavioral Impacts

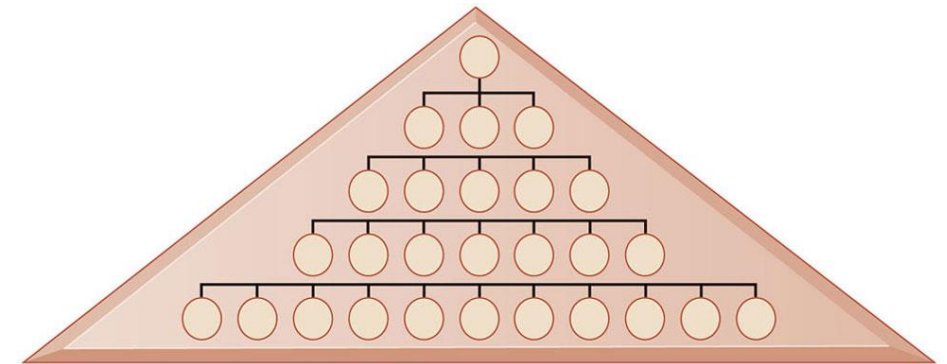
How IS transforms work behaviors and decision-making processes

- **IT Flattens Organizations:**

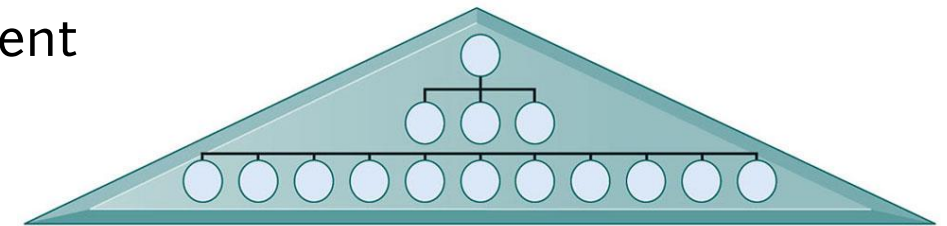
- Decision making is pushed to lower levels
- Fewer managers are needed (IT enables faster decision making and increases span of control)
- E.g., large retail chains empowering store-level employees with information access for independent decision-making

- **Postindustrial organizations**

- Organizations flatten because in postindustrial societies, authority increasingly relies on knowledge and competence rather than formal positions



A traditional hierarchical organization with many levels of management



An organization that has been "flattened" by removing layers of management

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# Understanding Organizational Resistance to Change

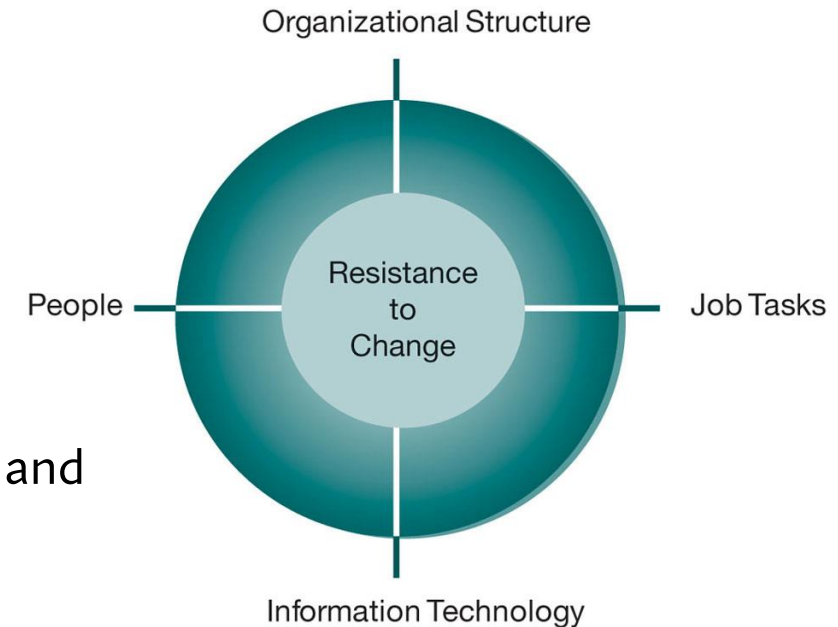
- Information systems become bound up in organizational politics because they influence access to a key resource—information
- Information systems potentially change an organization's structure, culture, politics, and work
- Resistance to IT innovation centers around four key factors:
  - Nature of the innovation
  - Structure of organization
  - Culture of organization
  - Tasks affected by innovation

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# Implications for IS Design

To deliver genuine benefits, information systems must be built with a clear understanding of their organizational context.

- Information systems potentially change an organization's structure, culture, politics, and work
- When planning a new system, six key factors must be evaluated:
  - The environment in which the organization must function
  - Organizational structure: hierarchy, specialization, routines, and business processes
  - The organization's culture and politics
  - The type of organization and its style of leadership
  - Principal interest groups affected and worker attitudes towards the system
  - Tasks, decisions, and business processes the system will support



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# Strategic Framework

In almost every industry, some firms perform better than others through effective use of information systems.

Key Strategic Models to understand how IS enables competitive advantage:

- Porter's competitive forces model
- The value chain model
- Synergies
- Core competencies
- Network economics

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# Using IT/IS to develop Competitive Advantages (CA)

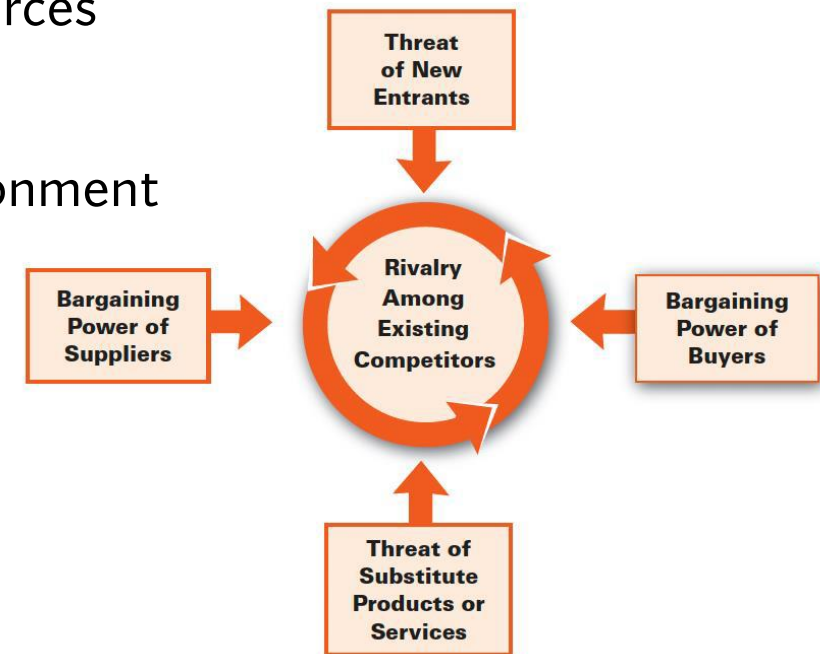
1. Identifying Where to Compete: Analyzing Competitive Forces
2. Identifying How to compete: Choosing a Generic Strategy
3. Identifying How to compete: Resources and capabilities

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# Using IT/IS to develop Competitive Advantages (CA)

## 1. Identifying Where to Compete: Analyzing Competitive Forces

- Why do some firms become leaders in their industry?
  - With limited resources, some organizations analyze the competitive forces within their industry to better understand where to focus their resources
- Michael Porter's competitive forces model
  - Provides general view of firm, its competitors, and environment
- Five competitive forces shape fate of firm:
  1. Rivalry among existing competitors
  2. Threat of new entrants
  3. Bargaining power of suppliers
  4. Bargaining power of buyers
  5. Threat of substitute products or services



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# Porter's Competitive Forces Model

## 1. Rivalry among existing competitors

- The intensity of rivalry is high if:
  - Competitors are numerous or are roughly equal in size and power
  - Industry growth is slow
  - Exit barriers are high
  - Rivals are highly committed to the business and have aspiration for leadership
  - Firms cannot read each other's signals well because of lack of familiarity with one another



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# Porter's Competitive Forces Model (cont.)

## 2. Threat of new entrants

- Barriers to entry:
  - Supply-side economics of scale
  - Demand-side benefits of scale
  - Customer switching costs
  - Capital requirements
  - Incumbency advantages independent of size
  - Unequal access to distribution channels
  - Restrictive government policy
- Expected retaliation
  - E.g., price war

Barriers to Entry in Different Market Structures

| Type of market structure | Level of barriers to entry              |
|--------------------------|-----------------------------------------|
| Perfect competition      | Zero barriers to entry                  |
| Monopolistic competition | Medium barriers to entry                |
| Oligopoly                | High barriers to entry                  |
| Monopoly                 | Very high to absolute barriers to entry |

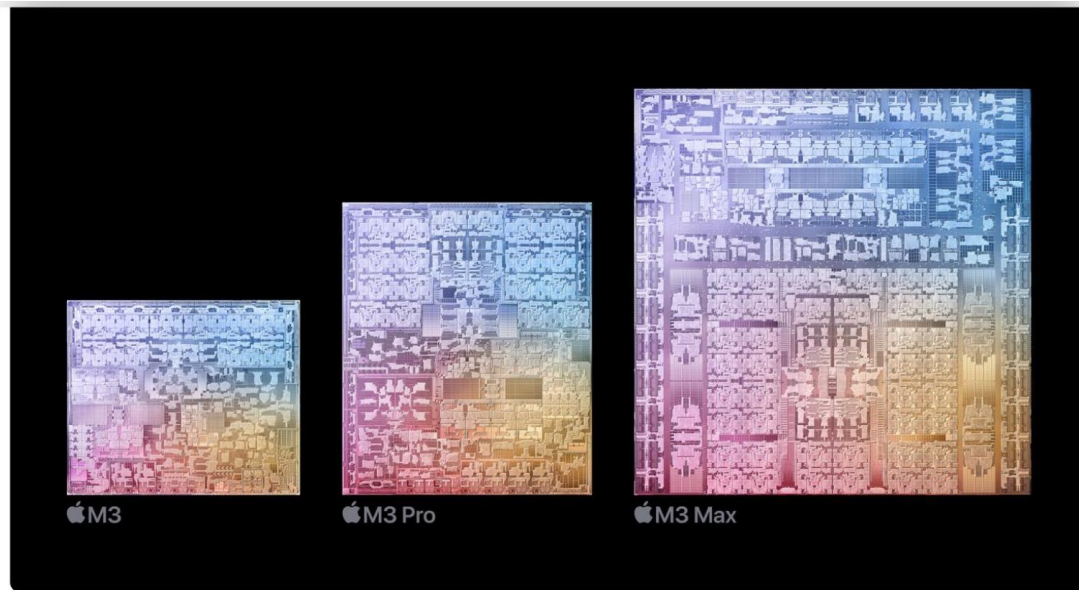
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# Porter's Competitive Forces Model (cont.)

## 3. Bargaining power of suppliers

- A supplier group is powerful if:
  - It is more concentrated than the industry it sells to.
  - The supplier group does not depend heavily on the industry for its revenue
  - Switching cost in changing suppliers
  - Suppliers offer products that are differentiated
  - No substitute for what the supplier group provides
  - The supplier group can credibly threaten to integrate forward into the industry.

# Discussion: Bargaining power of suppliers



The M3 family of chips is built using the industry-leading 3-nanometer process technology, and continues the tremendous pace of innovation in Apple silicon.

PRESS RELEASE

October 30, 2023

## Apple unveils M3, M3 Pro, and M3 Max, the most advanced chips for a personal computer

The industry's first 3-nanometer chips for a personal computer debut a next-generation GPU architecture and deliver dramatic performance improvements, a faster CPU and Neural Engine, and support for more unified memory

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# Discussion (cont.)

- Apple's M3 Chip Launch – Key Insights
- Apple unveiled M3, M3 Pro, and M3 Max, featuring 3nm technology.
- TSMC is the sole manufacturer, giving it strong leverage over Apple.
- Apple relies on TSMC's advanced manufacturing capabilities, increasing dependency.
- How This Relates to Supplier Power?
  - Differentiated Product by TSMC
  - High Switching Costs for Apple if moving to another supplier like Intel or Samsung.
  - Supplier Concentration as only few (TSMC, Samsung, Intel) can produce such advanced chips.
  - Threat of Forward Integration if TSMC started designing its own processors.
- **Question:** Can Apple reduce supplier power?

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# Porter's Competitive Forces Model (cont.)

## 4. Bargaining power of buyers

- A customer group has negotiation power if:
  - There are few buyers, or each one purchase in volumes that are large relative to the size of a single vendor.
  - The industry's products are standardized or undifferentiated.
  - Buyers face low switching costs in changing vendors.
  - Buyers can credibly threaten to integrate backward and produce the industry's product themselves if vendors are too profitable.
- E.g., Walmart's Bargaining Power Over Suppliers
  - Walmart is one of the world's largest retailers, purchasing huge volumes from suppliers
  - Walmart can easily switch suppliers if one doesn't meet cost expectations. Also has its own private-label brand (Great Value).

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# Porter's Competitive Forces Model (cont.)

## 5. Threat of substitute products or services

- A substitute performs the same or a similar function as an industry's product by a different means
- Threat of substitute products or services is high if:
  - It offers an attractive price-performance trade-off to the industry's product
  - The buyer's cost of switching to the substitute is low

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# Porter's Competitive Forces Model (cont.)

Shortcomings:

- Not very specific about 'what exactly to do' to gain CA
- Does not provide a methodology to achieve operational excellence
- Next week: The Value Chain Model

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# Self-Study

Please self read the next 2 slides



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# Using IT/IS to develop Competitive Advantages (CA)

## 2. Identifying How to compete: Choosing a Generic Strategy

- Low-cost leadership strategy
    - Lower price than competitors
    - Greater efficiency, decrease relevant costs involved
      - Avoid unprofitable customer accounts, minimize running cost or ancillary, such as R&D
  - Product differentiation
    - New products/services
    - Innovation
    - Involve high start-up and running cost
  - Focus on market niche
    - Specific market focus
  - Strengthen customer and supplier intimacy
    - Use information systems to develop strong ties and loyalty with customers and suppliers
    - Increase switching costs
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# Using IT/IS to develop Competitive Advantages (CA)

## 3. Identifying how to compete: Resources and capabilities

- Resources
  - The organization's specific assets that are utilized to achieve cost or product differentiation from its competitors.
  - E.g., proprietary technology, brand equity, a loyal and established customer base
- Capabilities
  - The organization's ability to leverage these resources in the marketplace.
- Distinctive competencies
  - E.g., innovation, agility, quality, or low cost

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Next

# The Business Value Chain Model